

Easy Visa Analysis

Python Foundations

UT Austin AI/ML Business Applications Course

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Executive Summary

Based off of our analysis of the data we have here at EasyVisa, here are some business insights and recommendations -

1. Prior Job Experience Increases Probability of Approval: Applicants with job experience had a 74% visa approval rate, compared to 56% for those without any job experience. We recommend that employers should prioritize candidates with prior work experience when it comes to sponsoring visas. EasyVisa can emphasize this factor in an internal pre-screening model to rank applicants.

2. Higher Education Greatly Boosts Certification Chances: Doctorates have a higher approval rate (~87%) than other degrees. We recommend when selecting candidates, to give weightage to Master's and Doctorate holders. We would also encourage skill-building programs for candidates with lower education levels to boost eligibility.

3. Higher Wage Correlates with Visa Success: Certified cases had significantly higher median wages than denied cases. The boxplots also showed wage to be a strong differentiator. We recommend that employers should ensure that offered salaries meet or exceed market expectations. EasyVisa's model can potentially flag low-wage cases for review or suggest appropriate wages.

Executive Summary

4. Unit of Wage: Annual salary is strongest indicator. We would recommend encouraging applicants and employers to use "Yearly" as the unit of wage reporting. Applications marked as "Hourly" can be reviewed or adjusted to improve approval chances.

5. Full-Time Positions Have Higher Approval: Full-time roles are significantly more likely to be certified than part-time positions. We recommend prioritizing full-time positions when preparing visa applications. EasyVisa could flag part-time applications as lower likelihood for success.

6. Geographic Differences Exist in Approval Rates: Certification rates vary across continents and U.S. regions. Applicants from Europe and North America fare better than others. Northeast and South regions of employment in the U.S. see more certified cases. We recommend performing further qualitative analysis into why certain continents or regions face higher denial rates. For employers, consider location-based best practices when submitting applications.

7. Job Training Requirement Negatively Affects Approval: Applicants requiring job training have lower success rates. I would recommend, if possible, presenting the candidate as fully trained or pre-qualified. For applicants needing training, bolster applications with extra documentation or credentials.

Executive Summary

8. Company Size is Not a Strong Predictor, but Very Small or Very Large Firms Show Variance: Though not the top driver, very small or very large companies showed variability in outcomes. I would recommend employers with extreme company sizes to ensure compliance documentation is tight. EasyVisa can consider integrating employer credibility scores in future model iterations.

9. Data Strategy Recommendation: Address Class Imbalance: Original dataset is imbalanced (66.8% certified vs. 33.2% denied). Oversampling improved F1 scores significantly. I recommend retaining SMOTE or similar sampling techniques in production models and regularly monitoring model drift or change in base rate of certified/denied cases.

Business Problem Overview and Solution Approach

The U.S. Office of Foreign Labor Certification (OFLC) receives hundreds of thousands of visa applications annually from employers seeking to hire foreign workers. Each application must be carefully reviewed to determine whether it should be **certified or denied**, ensuring compliance with labor laws and protection of local workers' wages and opportunities.

This **manual review process is time-consuming, inefficient, and prone to bottlenecks**, especially as the number of applications increases yearly. The current process lacks predictive intelligence to help assess visa approval likelihood upfront.

To assist OFLC in automating and enhancing the efficiency of the visa evaluation process by building a machine learning (ML) model that can:

- Predict the likelihood of visa **certification vs. denial**,
- Identify **key factors driving approvals**,
- Help employers and applicants understand **how to strengthen their applications**.

Business Problem Overview and Solution Approach

To address this business problem, we used the EasyVisa data set to do the following -

1. Data Understanding & Exploration
2. Data Preprocessing
3. Model Building & Evaluation
4. Hyperparameter Tuning
5. Final Model Selection
6. Actionable Insights & Recommendations

EDA Results

Key Findings –

Education: Doctorate and Master's holders have the highest approval rates.

Experience: Applicants with job experience had a ~74% approval rate, compared to ~56% for those without.

Training Requirement: Applicants requiring training are less likely to be approved.

Wage Unit: Applications with wages reported yearly had the highest approval rate.

Prevailing Wage: Higher wages correlate with higher certification probability.

Continent: Asia dominated the applications and had moderate success. Africa and Oceania had the lowest success rates.

EDA Results

Does higher education increase the chances of visa certification for well-paid jobs abroad?

Yes – Higher education does increase the chances of visa certification, especially in combination with:
Higher prevailing wages, Relevant job experience, Full-time positions.

How does visa status vary across different continents?

Europe and Africa had the highest approval rates (~79% and ~72%).
Asia had the most applications and a solid ~65% approval rate.
South America had the lowest success rate (~58%)

Does having prior work experience influence the chances of visa certification for career opportunities abroad?

Applicants with prior work experience are significantly more likely to be certified.

Is the prevailing wage consistent across all regions of the US?

Prevailing wages vary significantly across U.S. regions (e.g., Northeast, South, Midwest, West, Island).
Some regions (e.g., Northeast and West) offer higher median wages than others.

Does visa status vary with changes in the prevailing wage set to protect both local talent and foreign workers?

Certified applicants tend to have higher average prevailing wages. Distribution plots show that denied applications cluster around lower wage values.

Does the unit of prevailing wage (Hourly, Weekly, etc.) have any impact on the likelihood of visa application certification?

Applications with yearly wages have far higher approval rates. Hourly wages are associated with the lowest approval success

Data Preprocessing

Duplicate Value Check -

using `data.duplicated()` we found that there were no duplicate rows found in the dataset (25,480 records).

Missing Value Detection and Treatment -

using `data.info()` we found that all columns had 25,480 non-null values.

Outlier Detection and Treatment - Boxplots revealed outliers in: `no_of_employees` (some values > 600,000) as well as `prevailing_wage` (max = \$319,210). We found 33 negative values in `no_of_employees`. Negative values in `no_of_employees` were replaced using absolute value

Feature Engineering - Target variable `case_status` was in text format: "**Certified**" or "**Denied**".

Multiple categorical variables needed transformation. Converting the target into numeric format was necessary for ML models. Dropping `case_id` avoided noise and ensures the model learns from meaningful variables.

Data Preparation for Modeling -

A. Splitting the Data: Dataset was imbalanced (~67% certified, 33% denied). Stratification ensured class distribution remains consistent across all sets. A small test set (3%) is retained for final unbiased performance check.

B. Encoding Categorical Variables: Columns like `continent`, `education_of_employee`, `unit_of_wage`, etc., were categorical. ML models like decision trees and boosting algorithms require numerical input.

Model Performance Summary and Final Model Selection

After training and tuning models using RandomizedSearchCV, the following three models were evaluated:

- *Gradient Boosting (GBM)*
- *AdaBoost*
- *Random Forest*

All models performed well, but **GBM** slightly outperformed the others in terms of F1 Score and generalization. Oversampling (SMOTE) produced better results than undersampling due to retention of full information.

The Final Model: Gradient Boosting Classifier

This model was chosen for its balanced precision, recall, and F1-score on the validation set, and consistent performance across CV folds.

Model Performance Summary and Final Model Selection

Performance of the Final Model

F1 Score - 0.8125

Strengths:

- High recall means it correctly identifies most cases that should be certified.
- Strong F1 score indicates a great balance between precision (avoiding false positives) and recall (avoiding false negatives).
- Handles class imbalance effectively due to oversampling and model tuning.

Most Important Factors Used by the ML Model:

Tree-based models like GBM provide feature importance scores.

Below are the top contributors:

- prevailing_wage, education_of_employee, has_job_experience, unit_of_wage, full_time_position, region_of_employment, requires_job_training, no_of_employees

Applicants with higher wages, higher education, and job experience are significantly more likely to have their visas certified. Administrative factors like job type and region also play supporting roles.

Data Background and Contents

The dataset is sourced from the U.S. Office of Foreign Labor Certification (OFLC), which manages labor certification processes for employers seeking to hire foreign workers under temporary or permanent visa programs.

Total Records: 25,480

Total Features: 12 (case_id, continent, education_of_employee, has_job_experience, requires_job_training, no_of_employees, yr_of_estab, region_of_employment, prevailing_wage, unit_of_wage, full_time_position, case_status)

Model Building - Original Data

A variety of supervised classification models were trained on the original, imbalanced dataset (without SMOTE or undersampling) to predict visa application outcomes.

Evaluation Metric Focus:

F1 Score was the primary metric due to the class imbalance. Also considered: **Accuracy, Precision, and Recall.**

Gradient Boosting (GBM):

This was the best performer across all metrics. Strong F1 score (0.8196), meaning a great balance between precision and recall. There was low bias and good generalization, despite the imbalanced data. Selected for further tuning and testing.

AdaBoost and XGBoost:

Performed closely to GBM. Slightly lower precision and F1 scores but still strong contenders. XGBoost had the advantage of faster training but slightly lower performance.

Model Building - Original Data

Random Forest:

This model has strong recall but lower precision. May overfit slightly compared to boosting models.

Bagging and Decision Tree:

This model had significantly lower performance. This is not recommended as final models without resampling.

Conclusion:

Boosting algorithms (especially **Gradient Boosting**) are more robust on imbalanced data without needing immediate resampling. **GBM** was selected for tuning because it consistently produced the highest F1 score, indicating a strong trade-off between correctly predicting approvals and denials.

Model Building - Oversampled Data

Oversampling Method Used -

Technique: SMOTE (Synthetic Minority Over-sampling Technique)

SMOTE was applied to the training set only to address class imbalance by synthetically generating new samples of the minority class (Denied cases).

Gradient Boosting (GBM): This remained the best performer even after oversampling. F1 Score was 0.8125. The balanced recall (0.844) and precision (0.778), make it robust for real-world use. This was chosen as the final model for deployment.

AdaBoost: This is almost identical performance to GBM (F1: 0.8120). Slightly lower recall, but competitive precision.

XGBoost: This is very close third with an F1 Score of 0.8040. This is slightly lower generalization compared to GBM.

Random Forest: This is the lower F1 score due to reduced precision. This has strong recall, but may predict too many positives.

Bagging & Decision Tree: This performance improved with SMOTE, but still underperformed compared to ensemble boosting models.

Conclusion:

- Oversampling with SMOTE greatly improved recall and balanced the F1 score across all models.
- Gradient Boosting remained the top model, confirming its robustness across data variations.
- SMOTE should be retained in the pipeline when deploying on future imbalanced datasets.

Model Building - Undersampled Data

Technique: Random UnderSampling (RUS)

Random undersampling was applied to the training set only to address the class imbalance by randomly removing samples from the majority class (Certified cases).

Gradient Boosting (GBM): This once again delivered the highest F1 score (0.7609) among all models on the undersampled data.

There is a good trade-off between recall (0.783) and precision (0.740). It is robust even with reduced training data.

AdaBoost: This is nearly identical performance to GBM. Slightly lower recall and F1, but still efficient for small datasets.

XGBoost: Performed slightly worse than AdaBoost and GBM, possibly due to limited training data from undersampling.

Random Forest: Solid performance, but again slightly lagging behind boosting models. More prone to overfitting with smaller training sets.

Bagging & Decision Tree: Lowest performing models. Dropping too much data reduced their ability to generalize.

Conclusion:

- Undersampling reduces training time, but sacrifices model performance due to information loss.
- Gradient Boosting and AdaBoost still outperform others, proving their resilience.
- While undersampling is useful for testing and fast experimentation, it is less preferred for deployment than oversampling methods like SMOTE.

Model Improvement - Hyperparameter Tuning

Best Performing Models Selected for Tuning -

Model	Sampling Method	Reason for selection
Gradient Boosting	Oversampled	Consistently highest F1 score across all datasets
AdaBoost	Oversampled	Competitive F1 with simpler structure, fast training
Random Forest	Undersampled	Best performer in undersampled group, high recall

These models were chosen based on a combination of **F1 Score**, **consistency**, and **training stability**.

Tuning Method Used - RandomizedSearchCV

- Used for all three models to efficiently explore hyperparameter spaces.
- 5-fold cross-validation on the **training set**.
- Stratified sampling to preserve class balance.
- Scored using **F1 score** as the optimization metric.

Model Improvement - Hyperparameter Tuning

1. Gradient Boosting (Oversampled)

Best Parameters:

n_estimators: 200

learning_rate: 0.1

subsample: 1.0

max_features: None

2. AdaBoost (Oversampled)

Best Parameters:

n_estimators: 200

learning_rate: 1.0

3. Random Forest (Undersampled)

Best Parameters:

n_estimators: 150

max_depth: 20

max_features: 'sqrt'

min_samples_split: 5

Model Improvement - Hyperparameter Tuning

Tuned Model Performance -

Gradient Boosting (Oversampled)

This was the final chosen model due to highest F1 score and stable performance.

- Excellent balance of recall and precision.
- Good generalization on unseen data.

AdaBoost (Oversampled)

- Lightweight, fast training model.
- Near-identical performance to GBM but slightly less consistent in cross-validation.

Random Forest (Undersampled)

- Strong performance given reduced training data.
- Recall matched GBM, but slightly lower F1 score due to decreased precision.

To conclude,

- Gradient Boosting with SMOTE and tuned hyperparameters was selected as the final model.
- Hyperparameter tuning significantly improved model robustness and generalization.
- Boosting algorithms outperformed bagging and single-tree models in both raw and tuned forms.

Files used –

- EasyVisa.csv
- EasyVisa_Low_Code_Notebook.ipynb

APPENDIX



Happy Learning !

